Rooftop Solar Photovoltaic Performance Assessment: Phase 1

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ROOFTOP SOLAR PHOTOVOLTAIC PERFORMANCE ASSESSMENT: PHASE 1

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Project Reference: INIT_SOLAR_PH1 (Phase-1 Rooftop Solar PV Project)

Executing Departments: Department of Electronics & Telecommunication (DEP_EXTC) and Campus Facilities (DEP_ADMN).

Location: Vivekananda Education Society's Institute of Technology (VESIT Green Campus), Chembur, Mumbai.

1. SYSTEM ARCHITECTURE AND TECHNICAL SPECIFICATIONS

The Phase-1 Rooftop Solar Photovoltaic (PV) installation (INIT_SOLAR_PH1) was commissioned in November 2023 under net-metering regulations established by the Maharashtra Electricity Regulatory Commission (MERC) in partnership with the local utility distribution licensee (Adani Electricity Mumbai Limited / BEST). The grid-tied solar system comprises a rated DC capacity of 220 kWp utilizing Tier-1 high-efficiency monocrystalline silicon photovoltaic modules (400 Wp rated output per panel, 20.8% module conversion efficiency) coupled with commercial three-phase string solar inverters.

The 220 kWp total capacity is distributed across four institutional building terraces:

- Academic Block A (BLD_ACA): 80 kWp DC capacity (200 modules of 400 Wp), coupled with two 40 kW string inverters.

- Academic Block B (BLD_ACB): 70 kWp DC capacity (175 modules), coupled with one 50 kW and one 20 kW inverter.

- Administration Block (BLD_ADM): 40 kWp DC capacity (100 modules), connected to one 40 kW inverter.

- Library (BLD_LIB): 30 kWp DC capacity (75 modules), connected to one 30 kW inverter.

All arrays are mounted on elevated hot-dip galvanized structural steel pergolas tilted at a fixed 19° south-facing azimuth, optimized for Mumbai's geographic latitude (19.0760° N) to maximize annual solar irradiance capture while providing unimpeded pedestrian access on terrace roofs.

2. ANNUAL GENERATION PERFORMANCE AND SEASONAL DYNAMICS

Throughout the 12-month evaluation cycle from July 1, 2023, to June 30, 2024, the 220 kWp solar PV system generated a verified gross total of 312,400 kWh of clean electricity:

- Quarter 1 (July - September): Generated 52,600 kWh. The southwest monsoon created dense overcast cloud cover and heavy rainfall, causing monthly generation to drop to seasonal lows of 14,200 kWh in July and 15,800 kWh in August.

- Quarter 2 (October - December): Generated 82,400 kWh as clear post-monsoon skies and moderate ambient temperatures restored monthly yields to 27,200 kWh.

- Quarter 3 (January - March): Generated 93,600 kWh, benefiting from high atmospheric transparency and optimal solar irradiance in Chembur.

- Quarter 4 (April - June): Generated 83,800 kWh. Peak monthly generation occurred in April, reaching 32,800 kWh, before late-May pre-monsoon cloudiness emerged.

Building-level solar contributions were: BLD_ACA produced 113,600 kWh; BLD_ACB produced 99,400 kWh; BLD_ADM produced 56,800 kWh; and BLD_LIB generated 42,600 kWh. Specific plant performance ratio (PR) averaged 78.4% over the annual cycle, comfortably exceeding the guaranteed contractual threshold of 75.0%.

3. ENVIRONMENTAL DECARBONISATION AND FINANCIAL AUDIT

By generating 312,400 kWh of on-site solar power, the college avoided importing an equivalent volume of fossil-intensive grid electricity from the Western Regional Grid. Applying the Central Electricity Authority baseline grid factor EF_ELEC_GRID (0.82 kg CO2e/kWh), the Phase-1 solar installation prevented exactly 256,168 kg CO2e (256.17 metric tonnes CO2e) of indirect Scope 2 emissions in 2024.

Financial analysis confirms commercial viability:

- Total Displaced Grid Electricity Value: At an average institutional commercial tariff of INR 11.00 per kWh, annual gross electricity bill savings equaled INR 34,36,400 (34.36 Lakhs).

- Operation and Maintenance Costs: Annual preventive servicing, robotic cleaning, and inverter monitoring totaled INR 2,80,000.

- Net Annual Financial Savings: INR 31,56,400.

- Capital Payback Horizon: Against a turnkey installation expenditure of INR 1.32 Crores, the realized simple payback period is exactly 4.18 years, delivering robust return on investment while advancing institutional target TRG_RENEWABLE and supporting SDG 7, SDG 11, and SDG 13.

4. MAINTENANCE REGIME AND SOILING DEGRADATION ANALYSIS

Telemetry from IoT sub-meters installed under INIT_IOT_ENERGY_MGT revealed that urban particulate accumulation from Mumbai traffic and industrial activity in Chembur causes substantial soiling loss. Unwashed arrays experienced a measured 12.4% degradation in daily power yield over a three-week dry period in January 2024. Consequently, a bi-weekly automated water cleaning protocol utilizing harvested rainwater from INIT_RWH_SYSTEM was established, recovering 10.8% of lost yield. Recommendations emphasize executing the Phase-2 200 kWp expansion (INIT_SOLAR_PH2) across BLD_AUD and parking zones to achieve target TRG_RENEWABLE (50% clean electricity by 2030).

INSTITUTIONAL AUDIT & COMPLIANCE VERIFICATION

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